AP BIOLOGY TOPIC 4.3

Signal Transduction Pathways: Cellular Responses, Mutations, Agonists & Antagonists

UNIT 4: CELL COMMUNICATION & CELL CYCLE (10–15% EXAM WEIGHT)

IST-3.E | IST-3.F | IST-3.G | PATHWAY DISRUPTIONS

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 4.3 (SIGNAL TRANSDUCTION PATHWAYS: CELLULAR RESPONSES, MUTATIONS, AGONISTS & ANTAGONISTS)

CED Core Vocabulary

3. Apoptosis (Programmed Cell Death) — External death ligands or internal DNA/mitochondrial damage signals (cytochrome c release) activate a cascade of caspase proteases and nucleases that neatly...

1. Receptor Mutations — Extracellular Binding Domain Mutation: Ligand cannot bind → pathway remains permanently OFF even when hormone is present (e.g., androgen insensitivity or...

Constitutive Activation (Always "ON") — Mutant receptor fires continuously without any ligand bound → drives uncontrolled cell division (cancer , e.g., mutant HER2/EGF receptors).

4. Transcription Factor Mutations — Even if the receptor, second messengers, and kinases all work normally, a mutated transcription factor that cannot bind DNA causes the final gene expression...

Biological Roles — Embryonic development (removing webbing between human fingers/toes; tadpole tail resorption), eliminating virus-infected or DNA-damaged cells, and preventing...

Intracellular Domain Loss-of-Function — Ligand binds outside, but signal cannot be relayed inside.

3. Protein Kinase / Phosphatase Mutations — Inactive Kinase: Halts relay at that step; no downstream phosphorylation occurs.

§2. CELLULAR RESPONSES TO SIGNAL TRANSDUCTION

3 Major Outcomes

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

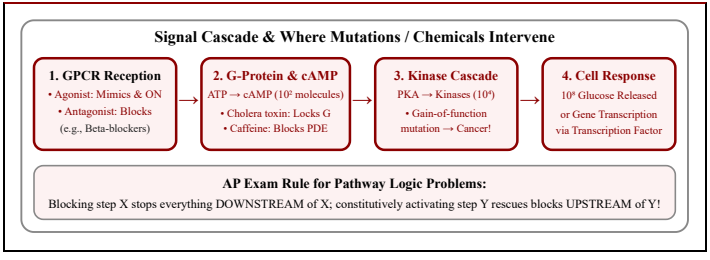
Cellular Responses to Signal Transduction ➔ **3 major functional outcomes** ➔ **3. Apoptosis (Programmed Cell Death)**

- Depending on the cell type, receptor, and intracellular relay proteins present, signal transduction pathways culminate in **3 major functional outcomes**:
- **1. Gene Expression Changes (Nuclear Response — Slower, Long-Term):**
  - Signaling cascades activate or inhibit specific **transcription factors** in the nucleus.
  - Turns transcription of target genes **ON or OFF** → alters mRNA and protein synthesis → changes cellular phenotype or triggers cell division/differentiation.
  - *Example:* **Steroid hormones** (cortisol, estrogen) or growth factors activating genes for cyclins.
- **2. Direct Changes in Cell Function / Metabolism (Cytoplasmic Response — Fast):**
  - Signals directly modify pre-existing cytoplasmic **enzymes, membrane transport proteins, ion channels, or cytoskeleton filaments** without altering DNA transcription.
  - *Examples:* **Insulin** triggers rapid exocytic insertion of **GLUT4 glucose transporters** into the plasma membrane to increase glucose uptake; **Epinephrine** activates glycogen phosphorylase to rapidly release glucose-1-phosphate.
- **3. Apoptosis (Programmed Cell Death):**
  - External death ligands or internal DNA/mitochondrial damage signals (cytochrome *c* release) activate a cascade of **caspase** proteases and nucleases that neatly dismantle the cell inside membrane blebs without causing inflammation.
  - **Biological Roles:** Embryonic development (removing webbing between human fingers/toes; tadpole tail resorption), eliminating virus-infected or DNA-damaged cells, and **preventing cancer**.

[COMMON AP EXAM FRQ TRAP] **Common Mistake:** Not all signaling pathways affect DNA or gene expression! Many pathways act **exclusively in the cytoplasm** (modifying existing enzymes or transporters like GLUT4) for rapid metabolic responses within seconds.

§3. PATHWAY AMPLIFICATION & DISRUPTION POINTS VISUALIZED

Epinephrine → cAMP → Glucose



§4. HOW MUTATIONS ALTER SIGNAL TRANSDUCTION PATHWAYS

Structure → Function

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

How Mutations Alter Signal Transduction Pathways ➔ **1. Receptor Mutations** ➔ **3. Protein Kinase / Phosphatase Mutations** ➔ **4. Transcription Factor Mutations**

- Mutations that alter the 3D structure of any protein in a pathway disrupt normal cellular responses:
- **1. Receptor Mutations:**
  - **Extracellular Binding Domain Mutation:** Ligand cannot bind → pathway remains permanently **OFF** even when hormone is present (e.g., androgen insensitivity or nephrogenic diabetes insipidus).
  - **Intracellular Domain Loss-of-Function:** Ligand binds outside, but signal cannot be relayed inside.
  - **Constitutive Activation (Always "ON"):** Mutant receptor fires continuously *without any ligand bound* → drives uncontrolled cell division (**cancer**, e.g., mutant HER2/EGF receptors).
- **2. Second Messenger Disruptions (cAMP , Ca^{2+}):**
  - Inactivating adenylyl cyclase → **no cAMP produced** → entire downstream cascade fails.
  - Blocking **cAMP** breakdown (inhibiting phosphodiesterase) → **prolongs and over-amplifies signaling**.
- **3. Protein Kinase / Phosphatase Mutations:**
  - **Inactive Kinase:** Halts relay at that step; no downstream phosphorylation occurs.
  - **Hyperactive / Constitutive Kinase (e.g., mutant Ras or B-Raf):** Continuously phosphorylates downstream targets even without upstream signal → linked to **uncontrolled mitosis / cancer**.
- **4. Transcription Factor Mutations:**
  - Even if the receptor, second messengers, and kinases all work normally, a mutated transcription factor that cannot bind DNA causes the **final gene expression response to fail** (e.g., mutant *p53* failing to activate cell-cycle arrest genes).

§5. CHEMICAL MODULATION: AGONISTS, ANTAGONISTS & TOXINS

Pharmacology & Pathogens

CHEMICAL TYPE	MECHANISM OF ACTION	HIGH-YIELD EXAMPLES
Agonists *(Activators)*	**Mimic the natural ligand**; bind to the receptor active site and induce the same conformational change to **activate the pathway**.	**Albuterol** (mimics epinephrine at lung β_2 receptors to dilate airways); **morphine/opioids** (endorphin agonists); synthetic birth control hormones.
Antagonists *(Blockers)*	**Bind to the receptor without activating it**; competitively block the natural ligand from binding → **inhibits pathway**.	**Beta-blockers** (block adrenaline/epinephrine receptors to lower heart rate & BP); **Narcan (naloxone)**; antihistamines.
Cascade Modulators & Toxins	Target intracellular G-proteins, adenylyl cyclase, phosphodiesterase (PDE), or kinases to block or lock signaling ON.	• **Caffeine:** Inhibits PDE → prevents **cAMP** breakdown → prolongs wakefulness signal. • **Cholera Toxin:** Modifies G-protein so it cannot hydrolyze GTP → locks adenylyl cyclase ON → massive **cAMP** → Cl^- & water efflux (severe diarrhea).

[CORE AXIOM / REMEMBER] **Agonist = Activator** (mimics ligand & turns receptor ON)
Antagonist = Blocker (occupies receptor & prevents natural ligand from activating it)

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 4.3 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 4.3: Signal Transduction Pathways: Cellular Responses, Mutations, Agonists & Antagonists ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 (Mitosis in Onion Root Tips & Lectin Signaling): Root tip squash stained with aceto-orcein/toluidine blue to compare mitotic index (% dividing cells = [cells in PMAT / total cells counted] × 100) between control meristem and phytohemagglutinin (lectin) or caffeine-treated roots. Chi-square (χ^2) tests whether observed mitotic vs. interphase counts differ significantly from control. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Mitotic Index = (N_mitosis / N_total) × 100. DNA Content (C) & Ploidy (n) tracking in diploid (2n) somatic cells: G_1 = 2n, 2C (unreplicated chromosomes); S phase = DNA synthesis (2n, 2C → 4C); G_2/Prophase/Metaphase = 2n, 4C (sister chromatids joined); Anaphase = transient 4n, 4C (chromatids separate into individual chromosomes); Post-Cytokinesis = 2n, 2C. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Signaling Disruption Case Study: Vibrio cholerae secretes Cholera Toxin, which enzymatically modifies the G_α subunit of intestinal GPCRs so it cannot hydrolyze GTP → GDP. Locked in its active GTP-bound state, G_α continuously activates adenylyl cyclase, causing runaway cAMP accumulation, constitutive PKA activation, CFTR Cl^-/H_2O efflux, and life-threatening secretory diarrhea. In cell cycle control, loss-of-function mutations in both alleles of TP53 ('guardian of the genome') prevent p21-mediated G_1/S arrest and apoptosis of DNA-damaged cells, driving oncogenic proliferation.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 4.3 to prove why the claim holds.

Support this site
Scan to donate